Dynamic report - simple discussion for the rocket's motion

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1. Introduction

In our dynamic class last week, the teacher illustrated a question which is about how to obtain the maximum velocity for an ideal rocket. In this part, atmospheric resistance and the variation of gravity are being ignored, and the final result would be very close to reality because consecutive incorrect iterations would produce a correct answer just by coincidence. As a member of engineer in the future, such inaccuracy and ignorance are not tolerated. Based on the reasons above, I have decided to analyze the racket with the mentioned factors(air drag and gravity) included and simulate the motion as realistically as possible, and obtain v(t).

2. Possibly involved forces in the system of rockets

In order to establish a desired and ideal environment of the system, I need to figure out what forces are involved and apply constraints to conduct the calculation. The following factors are forces we have to incorporate in real life.

a. Thrust
b. Gravity
c. Aerodynamic drag
d. Aerodynamic lift
e. Aerodynamic moments
f. Wind forces
g. Control System Forces
h. Internal Forces

With those forces, we can obtain the linear and rotational equations on the 2D plane by Newton's second law, ΣF = ma, to get:

$$m(t)v' = Ft + Faero + Fg + Fwind$$

$$I(t)\omega' + \omega \times (I(t)\omega) = Mthrust + Maero + Mrcs$$

For my research, my an ideal system only incorporates the effects of

Thrust, Gravity, and Aerodynamic drag without rotation. Therefore, I will presume the following constraints :

a. The rocket would vertically liftoff without flipping during the flight
   - Drop out aerodynamic lift and the possibility for the rocket to flip
b. The rocket is a rigid body
   - Drop out internal forces like damping and vibration
c. Angular Momentum of the rocket is steady, and no rotation occurs
   - So we can only discuss the linear equation without rotation involved
d. The Earth's rotation does not influence vertical energy
   - Ignoring the effects of Earth's rotation
e. Assume the rocket is shot in the desert
   - The desert is a desired spot for shooting a rocket since the wind force is slight. (The original question set the rocket at the North Pole, but actually, it was accompanied by extreme weather, which made it an inferior choice.)
f. Our system is ideal enough so that extra Control System forces are not necessary
   - Drop out Control System forces
g. Assume the environment is confined in the Troposphere(altitude < 11km), so we can apply some approximation for temperature and pressure.

With those assumptions, we have a simpler resulting equation:

$$m(t) * v' = Ft(m', v) + Fdrag(h) + Fg(h)$$

3. Detailed discussion for each force

a. Thurst

For Ft term, we can obtain thrust by the equation :

$$Ft = m' * v\,rel + (pe - p\infty) * Ae$$

Ft is the function that varies with mass rate(m'), related velocity of the nozzle(v rel), exit pressure(pe), ambient pressure(p∞), and Exit area of the nozzle(Ae). I assume our rocket is designed and can let pe equals p∞ , so we again simplify the equation to the following form :

$$Ft = m' * v\,rel$$

In general, those two factors can be controlled by engineers after the designation, so they would become parts of the parameters of the final v in the later simulation.

b. Aerodynamic drag

For the Fd term, we can obtain drag by the basic aerodynamic drag equation :

$$Fd = -\frac{1}{2}\rho(h) * Cd * A * v * |\,v\,|$$

We'll apply **International Standard Atmosphere (ISA)** model to get ρ(h) which look like:

$$\rho(h) = \frac{p(h)}{R * T(h)}, R = 287.05\,J/(kg \cdot K)$$

Based on that, Fd would become a value that varies with altitude(h) and velocity v. As the terms like T and p, we can approximate the value with these ISA models:

$$T(h) = T0 - L * h, \quad (\mathrm{K, Troposphere\ only})$$

$$p(h) = p0 * \left(1 - \frac{L*h}{T0}\right)^{\frac{g}{R*L}}, \quad (\mathrm{Pa, valid\ from\ 0\ to\ 86km\ altitude})$$

$$p0 = 101325\,Pa, g = 9.80665\frac{m}{s^2}, T0 = 288.15K, L = 0.0065K/m$$

First line is **Linear Lapse Rate Temperature Equation**, second is **Barometric Pressure Equation with Linear Temperature Gradient**, both of them should be apply when the altitude is less than 11km(within Troposphere). The unit of altitude h is meter.

In addition, when the rocket arrives Isothermal layer(12km < h < 20km), conditions of the atmosphere change. Hence, we apply other ISA models to predict temperature and atmospheric pressure:

$T(h) = Tb$, (K, Isothermal layer only)

$p(h) = pb * \exp\left(-\frac{g0*(h-11000)}{Tb*R}\right)$, (Pa, Isothermal layer only)

$Tb\ is\ the\ base\ temperature\ at\ altitude\ 11km, pb\ act\ as\ the\ same$

. As an observation, we can find out that the T only varies with the L term. Applying different models and L to get the proper T and p.

c. Gravity

For the term of Fg, we get the value of Fg by only mass and gravity g :

$$Fg = m(t) * g(h)$$

We would not neglect the influence with the slight variation of h, so g becomes a function :

$$g(h) = g0 * \frac{R^2}{(R+h)^2}$$

Furthermore, we know R is not the constant from place to place, so I'll use WGS-84 ellipsoid model to approximate value R :

$$R(\varphi)^2 = \frac{a^4 * (cos\varphi)^2 + b^4 * (sin\varphi)^2}{a^2 * (cos\varphi)^2 + b^2 * (sin\varphi)^2}$$

$$a = 6378137.0\ m, b = 6356752.314245\text{m}$$

I would like to pick the Mojave Desert at 35.0478° N as the shooting spot with fewer effects from the wind, so the corresponding R is roughly 6371124.526m

4. **Program logic and choose sample**

By combining the expanded equation above, we can get a complete equation:

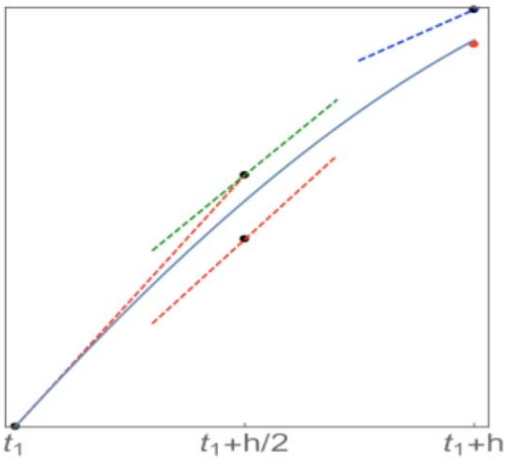
$$m(t) * \frac{dv}{dt} = -m'(t) * u - \frac{1}{2} * \rho(h) * Cd * A * v * | v | - m(t) * g(h).$$

To simulate this motion, it is necessary to simplify this equation and sort out the relationship between altitude, mass, velocity, and time into this form:

$$\frac{dv}{dt} = \frac{T - D - mg}{m}, \qquad \frac{dh}{dt} = v, \qquad \frac{dm}{dt} = -m'$$

It is clear first equation is a nonlinearly ODE, and we can only computer-simulate to solve v by iteration, so I chose Matplotlib as my simulation tool and used the mathematical relationship to get the function v(t). To get v(t), I have to make further constraints to set the gas relative velocity(v rel) and the increase rate of mass(m’) as constant. The code follows the key points below:

a. Define physical parameters and functions for repeated calls
b. Apply the RK4 method
c. Access to Matplotlib’s method to plot the v(t) graphic



$$k_1 = hf(y_1, t_1),$$
$$k_2 = h(f(y_1 + k_1/2, t_1 + h/2)$$
$$k_3 = h(f(y_1 + k_2/2, t_1 + h/2)$$
$$k_4 = h(f(y_1 + k_3, t_1 + h)$$
$$y_2 = y_1 + (k_1 + 2k_2 + 2k_3 + k_4)/6$$

Picture of RK4 logic

For the rocket parameters, I would like to select real rocket data to determine them. From that point, I found an original rocket with a simple design, **Black Brant V**, which partially satisfies my assumption rocket. Here is the basic information about this prototype. For the Drag Coefficient, unfortunately, I don’t have appropriate experimental data of Fd, thus I can only assign a reasonable number of Cd by its shape.

Diameter : 0.44 m

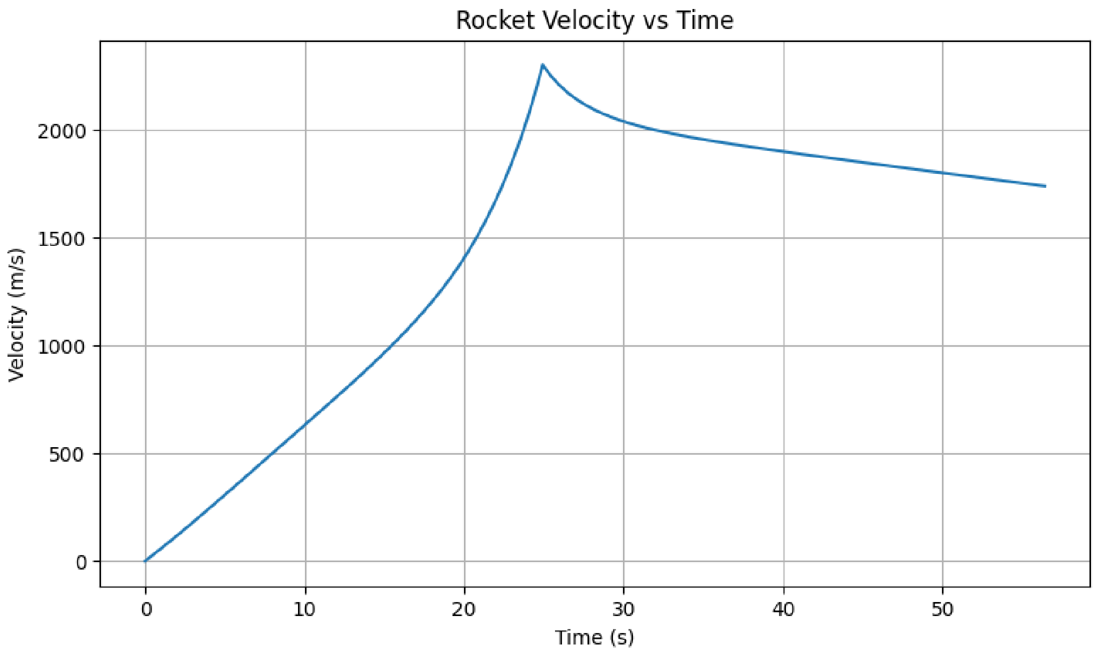
Mass at launch : 1,197 kg

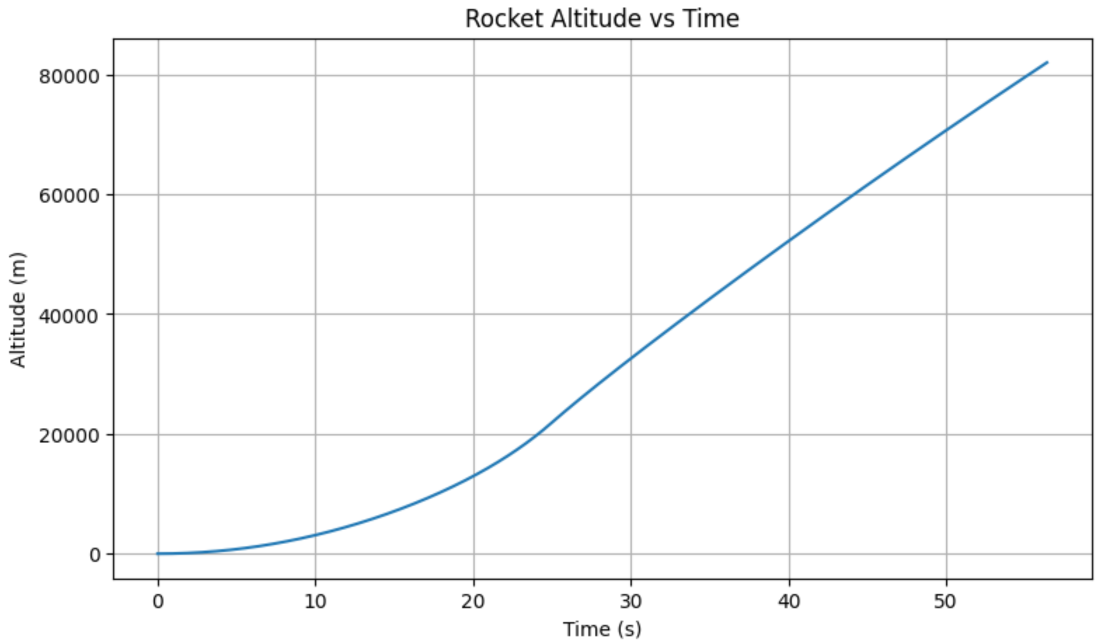
Drag Coefficient : 0.75 (approximation)

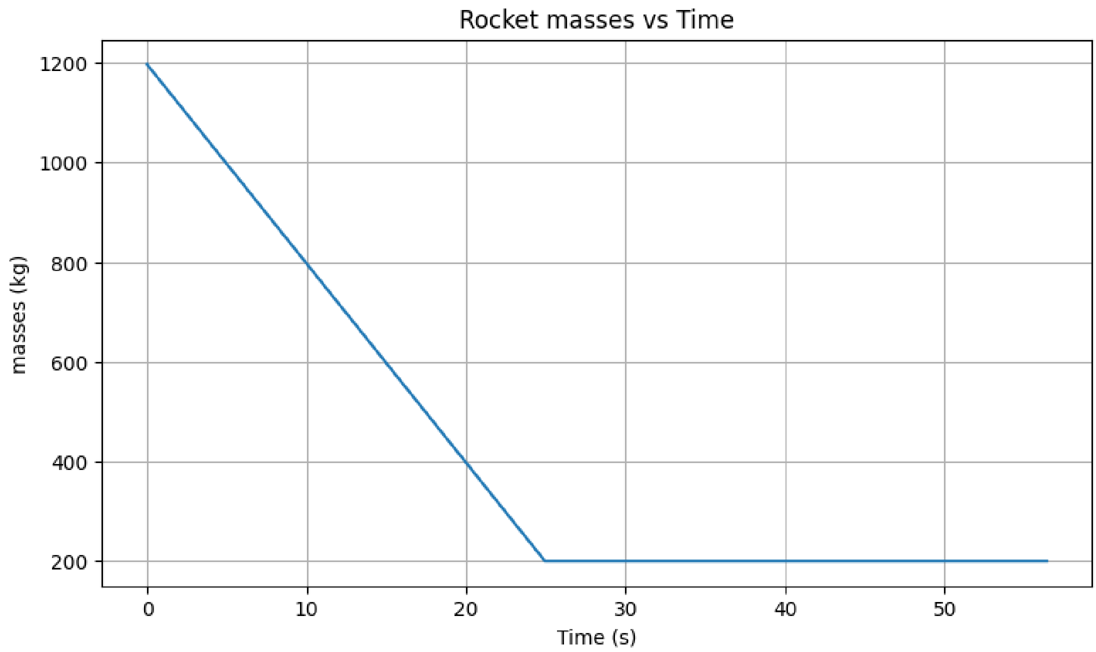
Picture of Black Brant VC

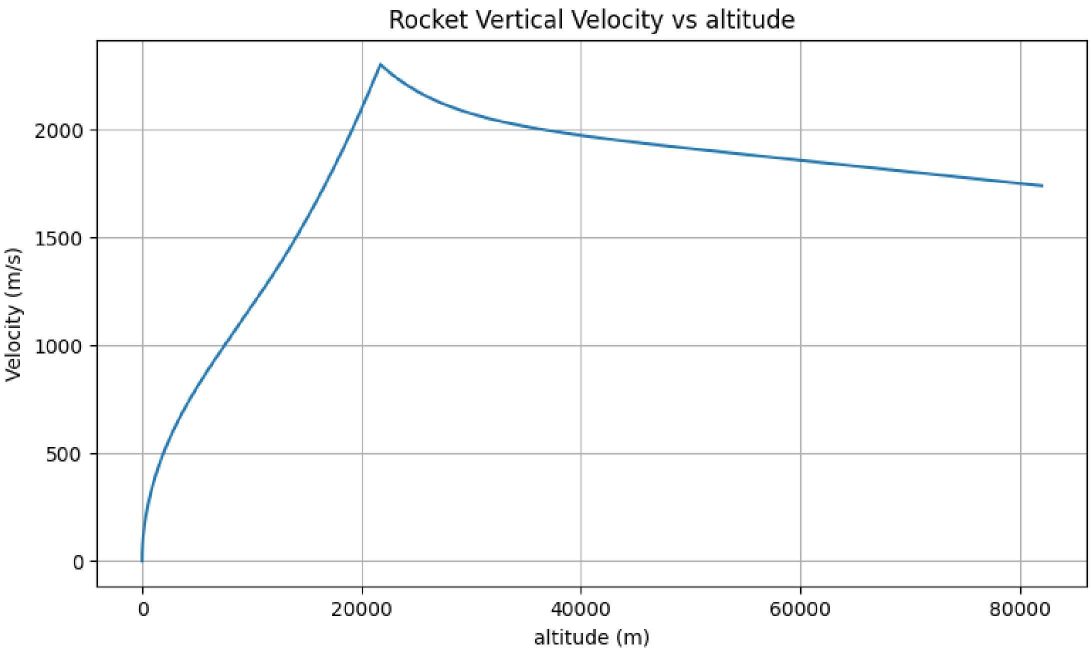
5. **Program and resulting graphic**

   I've stored my program on GitHub. Please go to the bottom of the pages of the reference part to access it. The graphics below are the simulation based on the mathematical models we apply:




Rocket Altitude vs Time
80000
60000
40000
20000
0
Altitude (m)
0
10
20
30
40
50
Time (s)


Rocket masses vs Time
1200
1000
800
600
400
200
masses (kg)
0
10
20
30
40
50
Time (s)



As we can see, the velocity of the rocket rapidly increases, but drops when the fuel runs out. Furthermore, as expected, the values of masses, velocity, and altitude are dependent, which is a fairly desirable result.

6. Conclusion

With my complete simulation, I'm able to predict various cases of the rocket with different parameters, and my future target is to expand my research by incorporating more effects I neglected at the beginning. Besides, I learned a lot from this analysis, such as the practical method to simulate the first-order derivative equation and how to select a proper formula for different conditions. I hope this experience can help me develop models and assist me in researching more complicated calculations.

7. Reference

Program on GitHub

→https://github.com/Mill1781/indie-project.git

Black Brant rocket info :

→https://en.wikipedia.org/wiki/Black_Brant_%28rocket%29?utm_source=chatgpt.com

Method to estimate Drag coefficients :

→https://www.researchgate.net/publication/330684511_Shape_effects_on_drag

→ https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/shape-effects-on-drag/?utm_source=chatgpt.com

Method of RK4 reference

→ https://www.youtube.com/watch?v=dShtlMl69kY&t=603s

Formula for temperature and pressure

→ https://www.aviationhunt.com/standard-atmosphere-calculator/

→https://en.wikipedia.org/wiki/International_Standard_Atmosphere

→https://en.wikipedia.org/wiki/Barometric_formula